Module I:
Basics of R programming

Module I: Basics of R programming

- Overview and installation
- Data types in R
- Input and output
- Subsetting
- Using **dplyr** to manipulate datasets
- Control structures
- R functions
- Simulations

Module I: Reference and Resource

- Reference: Roger Peng (2020), *R Programming for Data Science*, 5th Edition, lulu.com.

  The electronic version of the book can be downloaded freely from: https://leanpub.com/rprogramming

- The course materials were partially adopted from a course based on the above reference taught at John Hopkins University with the permission from Dr. Roger Peng.

- Other publicly available online tutorial and videos.

History and Overview of R

What is R?

- R is a comprehensive statistical environment and programming language for professional data analysis and graphical display.

- R is a “dialect” of the S language. It is a GNU project which implements the S language that was developed at Bell Laboratories. (The GNU project is a mass collaborative initiative for the development of free software, which was founded by Richard Stallman in 1978 at MIT.)

History of S/R

- S is a language that was developed by John Chambers and others at Bell Labs.
- S was initiated in 1976 as an internal statistical analysis environment - originally implemented as Fortran libraries.
- Early versions of the S language did not contain functions for statistical modeling.
- In 1988 the system was rewritten in C and began to resemble the system that we have today (this was Version 3 of the language).
- Version 4 of the S language was released in 1998 and is the version we use today.

Philosophy of S/R

In “Stages in the Evolution of S”, John Chambers writes:

“[W]e wanted users to be able to begin in an interactive environment, where they did not consciously think of themselves as programming. Then as their needs became clearer and their sophistication increased, they should be able to slide gradually into programming, when the language and system aspects would become more important.”

http://www.stat.bell-labs.com/S/history.html

Back to R

- 1991: Created in New Zealand by Ross Ihaka and Robert Gentleman. Their experience developing R is documented in a 1996 JCGS paper.
- 1993: First announcement of R to the public.
- 1995: Martin Mächler convinces Ross and Robert to use the GNU General Public License to make R free software.
- 1996: A public mailing list is created (R-help and R-devel)
- 1997: The R Core Group is formed (containing some people associated with S-PLUS). The core group controls the source code for R.
- 2000: R version 1.0.0 is released.
- Current R version 4.5.1 is released on 2025-06-13.

Features of R

- Syntax is very similar to S, making it easy for S-PLUS users to switch over.
- Runs on almost any standard computing platform/OS (Windows, macOS and Linux).
- Frequent releases (annual + bugfix releases); active development.
- Quite lean, as far as software goes; functionality is divided into modular packages.
- Graphics capabilities very sophisticated and better than most statistical packages.
- Useful for interactive work, but contains a powerful programming language for developing new tools (user -> programmer).
- Very active and vibrant user community; R-help and R-devel mailing lists and Stack Overflow

Features of R (cont'd)

- New statistical methods are usually first implemented in R.
- It's free!
- Able to access to the source code of any package built upon R.
- With free software, you are granted

- The freedom to run the program, for any purpose (freedom 0).

- The freedom to study how the program works, and adapt it to your needs (freedom 1).
- The freedom to redistribute copies so you can help your neighbor (freedom 2).

- The freedom to improve the program, and release your improvements to the public, so that the whole community benefits (freedom 3).

Drawbacks of R

- Essentially based on 40 year old technology.
- Little built in support for dynamic or 3-D graphics (but things have improved greatly since the "old days").
- "Free" means using the packages at your own risk. Also, no one is committed to implement your favorite method or debug for you.
- Objects must generally be stored in physical memory, so R is not memory efficient; but there have been advancements to deal with this too.
- Not ideal for all possible situations (but this is a drawback of all software packages).

An R community blog edited by RStudio
Boston, MA

200 POSTS | 158 TAGS

How to Teach R: Common mistakes

2017-02-22

by Garrett Grolemund

Would you like to teach people to use R? If so, I would like to jump-start your efforts.

I'm one half of RStudio's education team, and I've taught thousands of people to use R, usually in face-to-face workshops. Over time, I've come to appreciate that teaching R in a short workshop is an unusual challenge that requires an unusual approach: you cannot teach a short workshop in the same way that you would teach a college course, and you should not teach R in the same way you would teach Python, UNIX or C.

In the next few blog posts, I'll share the pedagogy that I've adopted for teaching R workshops. These ideas have made my life easier and my students happier (based on student feedback). I think they can do the same for you.

We'll begin in this post by identifying common mistakes that ensnare new R teachers. Each of these mistakes seems like a good idea at first glance, but leads to an unsuccessful short workshop, and I'll tell you why. To make things simple, I've recast each mistake as a principle to follow. Let's examine them one by one:

- **DO NOT teach R as if it were a programming language.** Why not? Because R is a programming language *for doing data science*. You can be confident that your students want to use R to make graphs, fit models, and impress their colleagues. Show them how to do these empowering things and then teach programming later, as a way to do these things even better. To be honest, if your students only wanted to learn how to program, they would be studying another language.

Design of the R System

- The R system is divided into 2 conceptual parts:

1. The "base" R system that you download from CRAN (https://cran.r-project.org/) and installed in your computer. CRAN is a network of ftp and web servers around the world that store identical, up-to-date, versions of code and documentation for R.

2. Everything else – all other packages contributed by users.

Design of the R System – the "base" system

- The "base" R system contains, among other things, the **base** package which is required to run R and contains the most fundamental functions.
- The other packages contained in the "base" system include **utils**, **stats**, **datasets**, **graphics**, **grDevices**, **grid**, **methods**, **tools**, **parallel**, **compiler**, **splines**, **tcltk**, **stats4**.
- There are also "Recommend" packages: **boot**, **class**, **cluster**, **codetools**, **foreign**, **KernSmooth**, **lattice**, **mgcv**, **nlme**, **rpart**, **survival**, **MASS**, **spatial**, **nnet**, **Matrix**.
- Precompiled binary distributions of the base system for major platforms (Linux, macOS and Windows) are maintained and updated and can be downloaded from CRAN (https://cran.r-project.org/).

Design of the R System – other packages

- CRAN: There are about 4000 packages on CRAN that have been developed by users and programmers around the world.
- The Bioconductor project (https://bioconductor.org) provide many package for the analysis of high-throughput genomic data and bioinformatics.
- People also often make packages available on GitHub (https://github.com/) and their personal websites. There is no reliable way to keep track of how many packages are available in this fashion.

References and Resources for R

- Tons of texts and tutorials.
- Manuals and help page of specific R packages.
- A long but not complete list of books on R: https://www.r-project.org/doc/bib/R-books.html
- Major technical publishers like Springer, Chapman & Hall/CRC have entire series of books dedicated to using R in various applications. For example, Springer has a series of books called *Use R*!